ZD-OPX100 — Zero-Drift, Chopper-Stabilized Operational Amplifier

Synthetic test datasheet for KB ingestion pipeline validation

Parameter	Value	Condition
Offset Voltage (Vos)	1 uV typ, 5 uV max	TA = 25C
Offset Voltage Drift	0.005 uV/C	-40C to 125C
Input Voltage Noise Density	11 nV/sqrt(Hz)	f = 1kHz
1/f Noise Corner	Not applicable (chopped)	Auto-zero architecture
Gain Bandwidth Product	2.8 MHz	typical
Quiescent Current	1.2 mA per amplifier	typical
Supply Voltage Range	2.7V to 5.5V single supply	or +/-1.35V to +/-2.75V dual
CMRR	140 dB typical	DC
PSRR	140 dB typical	DC

1. General Description

The ZD-OPX100 is a zero-drift operational amplifier that uses a chopper stabilization technique to continuously cancel input offset voltage and 1/f noise. Unlike conventional auto-zero architectures that sample and hold, the chopper topology modulates the input signal above the flicker noise corner, amplifies it, and demodulates it back to baseband, resulting in extremely low offset voltage drift over temperature and time. This makes the device well suited for precision instrumentation amplifiers, current sense amplifiers, and ultra low noise current source designs where long term stability is critical.

2. Noise Performance

Input voltage noise density is specified at 11 nV per square root Hz at 1 kHz, with no 1/f noise corner due to the chopping architecture. When used in a current source feedback loop, the dominant noise contribution typically shifts to the sense resistor's Johnson noise and the reference voltage noise rather than the amplifier itself, provided the sense resistor value and bandwidth are chosen appropriately. Designers building ultra low noise current sources should budget total output current noise as the root sum square of amplifier voltage noise divided by the sense resistance, resistor thermal noise, and reference noise contributions.

3. Application: Precision Current Source

In a Howland or modified Howland current pump topology, the ZD-OPX100's extremely low offset voltage and drift directly translate into excellent current source accuracy and stability over temperature, since any amplifier input offset appears as a current error proportional to the sense resistor value. For best results, pair this amplifier with ultra-precision, low temperature coefficient resistors in the gain network to preserve the matching accuracy the topology depends on.

4. Power Supply Recommendations

The ZD-OPX100 operates from a single 2.7V to 5.5V supply or a dual symmetrical supply. For best noise and PSRR performance, supply rails should be locally decoupled with a 0.1uF ceramic capacitor placed close to the supply pins, in addition to a bulk 10uF capacitor per supply rail. When deriving bipolar rails from a single DC input, a low noise inverting charge pump or a small auxiliary LDO-based negative rail generator is recommended to avoid injecting switching noise into the signal path.